P-VTOL Control System Project Guide

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Introduction

This document serves as a practical guide for students working on the P-VTOL (Planar Vertical Take-Off and Landing) project.

The goal of the project is to design, model, control, and experimentally validate a one degree-of-freedom (1-DOF) aerial platform actuated by two brushless DC (BLDC) motors.

Mechanical Design

The project started with the design of a lightweight structure capable of rotating around a fixed pivot, parts stated below:

- Main beam
- Motor Holder ($\times 2$)
- Bearing Supports ($\times 2$)
- Sensor Mount ($\times 2$)
- Bearing Retainer ($\times 2$)
- **Base** (Material: Teflon, Dimensions: $30 \times 15 \times 3$ cm, Fabrication: CNC Machining)

The CAD model was developed using SolidWorks, as shown in Figure 1.

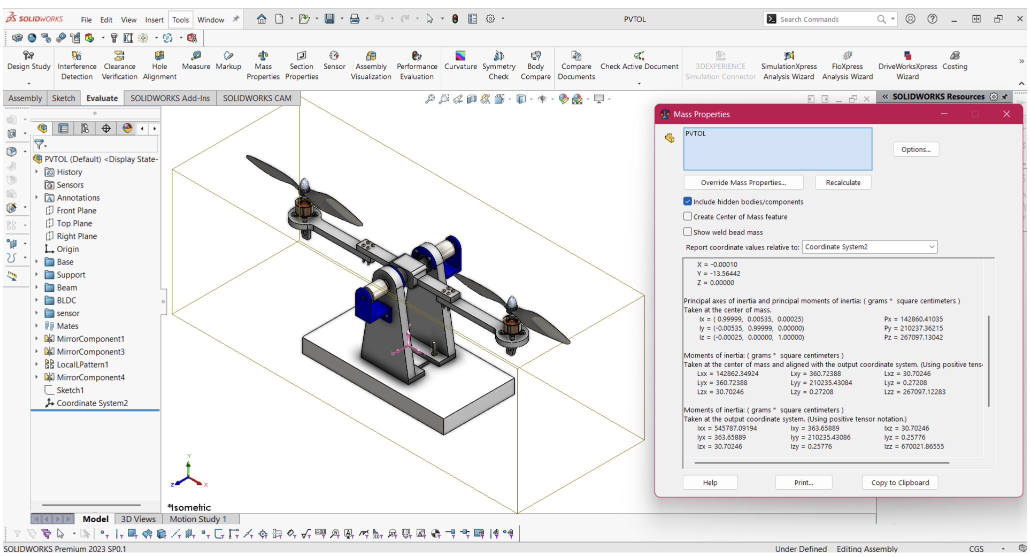

Figure 1: SolidWorks CAD Assembly Model

After finalizing the CAD design, the components were manufactured via 3D printing. The specific slicer configurations are outlined in Figure 2.

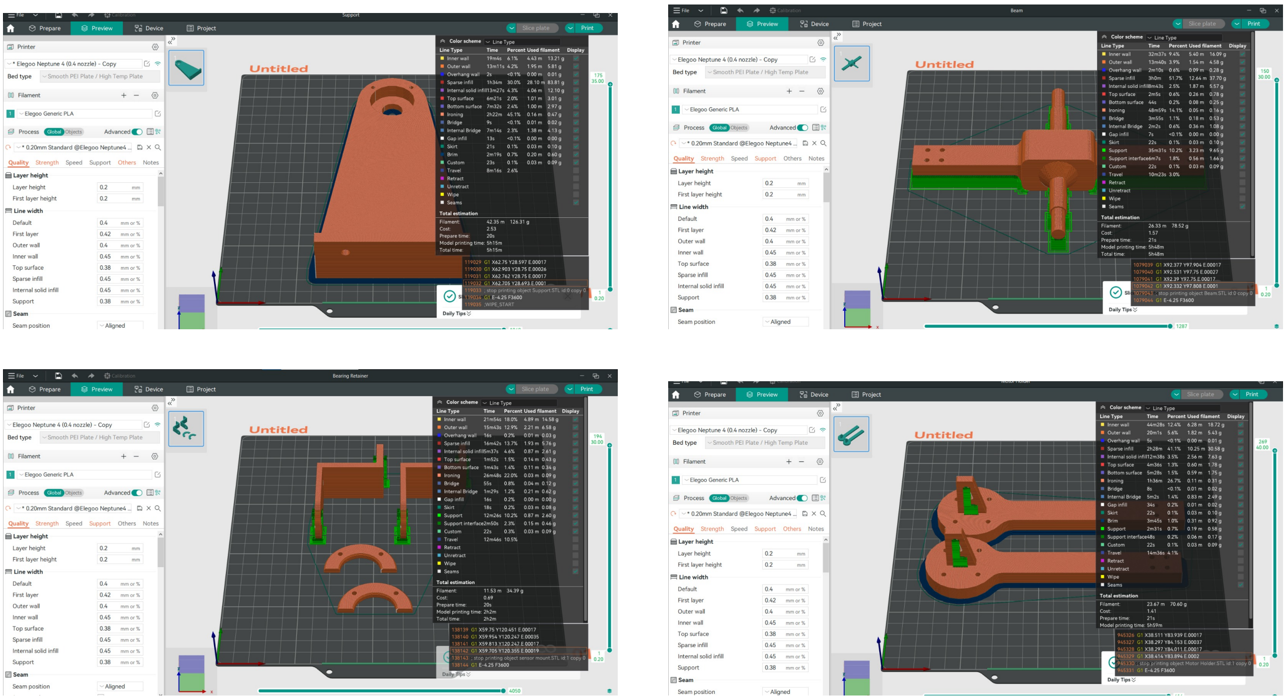

Figure 2: 3D Printing Settings

Printed elements were then assembled with the following mechanical components:

- M3, M4, M5 Nuts and Bolts
- 6201 Ball Bearings * 2
- Flexible Coupling 6.35*10 L30

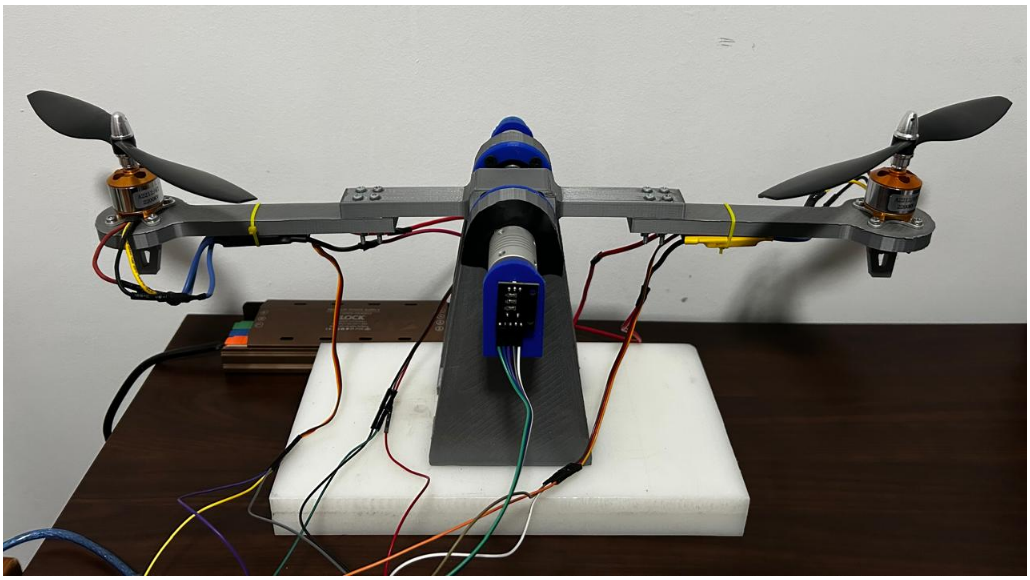

Figure 3: Final Assembly

Mathematical Modeling

The P-VTOL platform can be modeled as a rigid beam rotating about a fixed pivot. The angular position of the beam is denoted by θ.

Free Body Diagram

Figure 4 illustrates the free body diagram of the system.

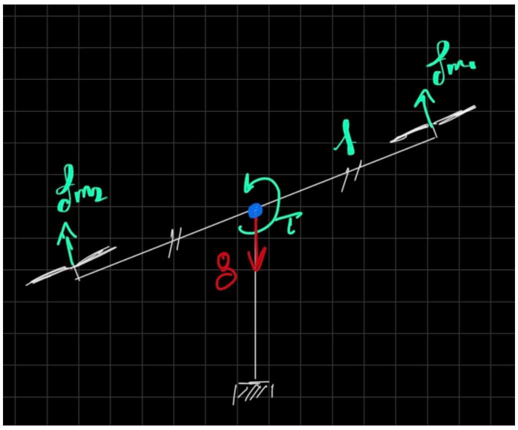

Figure 4: Free body diagram of the P-VTOL system

The control torque is generated by the differential thrust of the two motors,

$$\tau = l(F_R - F_L)$$

where l is the distance from the pivot to each motor.
To account for friction and aerodynamic losses, a viscous damping torque is included,

$$\tau_d = b\dot{\theta}$$

where b is the damping coefficient.

Euler-Lagrange Formulation

The kinetic energy of the system is

$$T = \frac{1}{2}J\dot{\theta}^2$$

where J is the moment of inertia about the pivot axis.
Since the beam is balanced around the pivot, the variation in potential energy is negligible,

$$V = 0$$

Therefore, the Lagrangian becomes

$$L = T - V = \frac{1}{2}J\dot{\theta}^2$$

Applying the Euler-Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = Q$$

$$Q = l(F_R - F_L) - b\dot{\theta}$$

$$J\ddot{\theta} + b\dot{\theta} = l(F_R - F_L)$$

$$u = l(F_R - F_L)$$

the equation of motion becomes

$$J\ddot{\theta} + b\dot{\theta} = u$$

State-Space Model

For controller design, the states are chosen as

$$x_1 = \theta, \qquad x_2 = \dot{\theta}$$

The state equations become

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{b}{J}x_2 + \frac{1}{J}u$$

which can be written in matrix form as

$$\dot{x} = Ax + Bu$$

where

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{b}{J} \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix}$$

The output vector is selected as

$$y = Cx + Du$$

with

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

‘‘‘

Measurements

The MPU6050 sensor was used to measure angle and angular velocity, The sensor was mounted on top of the beam center near the pivot point.

Angle Measurement

The accelerometer angle (Pitch Angle) was computed using

$$\theta_{acc} = \tan^{-1}\left(\frac{A_x}{A_z}\right) * 180/\pi$$

$$A_x = g\cos\theta \qquad A_z = g\sin\theta$$

Schematic for this equation is shown in Figure 5.

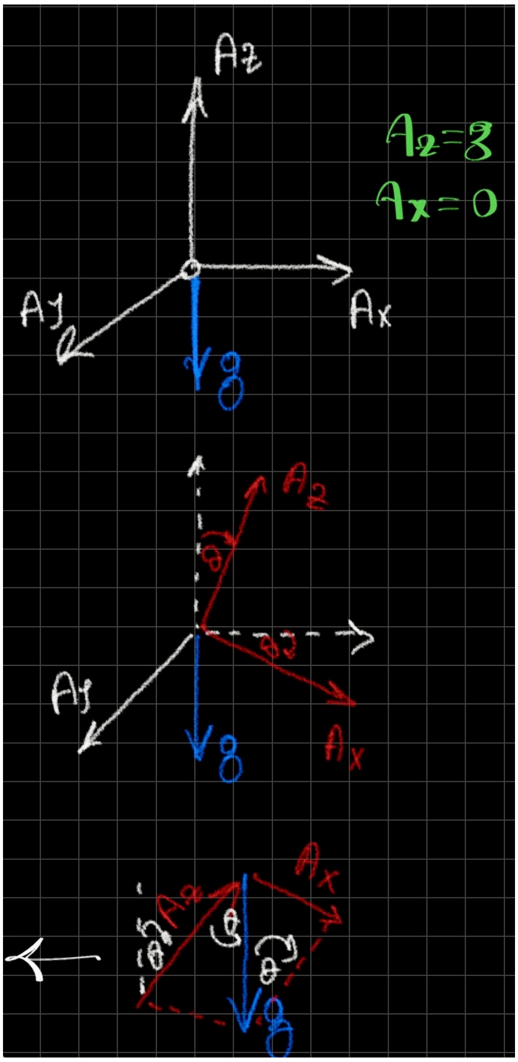

Figure 5: Obtaining Pitch Angle from Accelerometer

Angle Calibration

Measure angle at startup θ_{ref} then subtract it as in the equation below, this is to ensure that angle measured is calibrated at zero.

$$\theta = \theta_{acc} - \theta_{ref}$$

Angular Velocity Measurement

Angular velocity is defined as the rate of change of angular position over time:

$$\omega = \frac{\Delta\theta_p}{\Delta t} \quad \text{(rad/s)}$$

For the specific gyroscope sensor used, the sensitivity scale factor is given as 131 LSB $= 1°/\text{s}$. Therefore, the raw gyroscope reading in the y-axis (G_y) can be converted to degrees per second using:

$$\omega_{\text{deg/s}} = \frac{G_y}{131}$$

To ensure measurements are free from zero-rate drift or offset errors, a calibration value (G_0 or G_{off}) must be measured while the sensor is completely at rest. Subtracting this offset yields the corrected angular velocity in degrees per second:

$$\omega = \frac{G_y - G_0}{131}$$

To convert this corrected measurement into radians per second (rad/s), multiply it by $\frac{\pi}{180}$:

$$\omega_{\text{rad/s}} = \left(\frac{G_y - G_{\text{off}}}{131}\right) \times \frac{\pi}{180}$$

Angle Estimation and Sensor Fusion

Gyroscope Integration

The angular position θ can be measured by integrating the angular velocity ω over a time step dt using Euler's method:

$$\theta_{i+1} = \theta_i + \omega \cdot dt$$

Complementary Filter

To overcome individual sensor limitations, a complementary filter is implemented. It accounts for the low-frequency drift inherent in gyroscope integration while mitigating the high-frequency noise of the accelerometer data (θ_{acc}):

$$\theta = \alpha \cdot (\theta + \omega \cdot dt) + (1 - \alpha) \cdot \theta_{\text{acc}}$$

Where α is a tuning constant, typically chosen to be around 0.98.

System Parameters

The physical parameters of the system are defined as

$$l = 22 \text{ cm} = 0.22 \text{ m}$$

$$b = 0.02 \text{ N} \cdot \text{m} \cdot \text{s/rad}, \qquad J = 0.067 \text{ kg} \cdot \text{m}^2 \text{ (Obtained from SolidWorks)}$$

State-Space Model

Substituting the system parameters into the state-space equations yields the system matrices

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -0.3 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 15 \end{bmatrix}$$

The output and direct transmission matrices are selected as

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Control System Design

The characteristic equation of the system is determined by

$$\text{Ch. eqn.} \implies \det(sI - A) = 0$$

Evaluating the determinant of the state matrix gives

$$\begin{vmatrix} s & -1 \\ 0 & s+0.3 \end{vmatrix} = s(s+0.3) = 0$$

Solving for the roots yields the open-loop poles of the system at

$$s = 0, \qquad s = -0.3$$

Controllability Analysis

The controllability matrix M is constructed as

$$M = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 15 \\ 15 & -45 \end{bmatrix}$$

Evaluating the rank of the matrix yields

$$\text{Rank}(M) = 2 \qquad \text{"Controllable"}$$

The system is not in controllable canonical form, as second row of B is not 1;

State-Feedback Control

The state-feedback control law is defined as

$$u = -Kx = -\begin{bmatrix} K_1 & K_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Expressing this in terms of physical parameters θ and $\dot{\theta}$ gives

$$u = -K_\theta \theta - K_\omega \dot{\theta}$$

The closed-loop state equation is governed by

$$\dot{x} = (A - BK)x$$

Substituting the state matrix A, input matrix B, and gain vector K yields

$$\begin{bmatrix} 0 & 1 \\ 0 & -0.3 \end{bmatrix} - \begin{bmatrix} 0 \\ 15 \end{bmatrix} \begin{bmatrix} K_\theta & K_\omega \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 0 & -0.3 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 15K_\theta & 15K_\omega \end{bmatrix}$$

Consequently, the closed-loop system matrix $\tilde{A}$ is obtained as

$$\tilde{A} = \begin{bmatrix} 0 & 1 \\ -15K_\theta & -0.3 - 15K_\omega \end{bmatrix}$$

Closed-Loop Pole Placement

The characteristic equation of the closed-loop system is analyzed to determine the closed-loop poles

$$\det(sI - \tilde{A}) = 0$$

Substituting the identity matrix I and the closed-loop system matrix $\tilde{A}$ gives

$$\left| \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -15K_\theta & -0.3 - 15K_\omega \end{bmatrix} \right| = 0$$

Simplifying the matrix subtraction within the determinant yields

$$\begin{vmatrix} s & -1 \\ 15K_\theta & s + 0.3 + 15K_\omega \end{vmatrix} = 0$$

Computing the determinant results in the following polynomial expression

$$s(s + 0.3 + 15K_\omega) + 15K_\theta = 0$$

Expanding and grouping the terms by powers of s yields

$$s^2 + 0.3s + 15K_\omega s + 15K_\theta = 0$$

$$s^2 + (0.3 + 15K_\omega)s + 15K_\theta = 0$$

Controller Gain Calculation

The actual closed-loop characteristic equation is compared against the standard second-order system design template

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

The desired transient performance characteristics are specified as

$$\zeta = 0.7, \qquad \omega_n = 3.24$$

These parameters correspond to the desired closed-loop pole locations at

$$s_{1,2} = -2.25 \pm 2.33i$$

Equating the coefficients of the standard equation to the system's characteristic equation yields the design formulas

$$2\zeta\omega_n = 0.3 + 15K_\omega$$

$$\omega_n^2 = 15K_\theta$$

Solving this system of equations for the state-feedback gains gives

$$K_\theta = 0.7, \qquad K_\omega = 0.2824$$

Experimental Tuning

While the analytical design gives precise theoretical values, physical constraints or unmodeled dynamics often require empirical adjustment.

Comparing the calculated gains against experimental testing yields

$$K_\theta = 0.7, \qquad K_\omega = 0.2824$$

$$\text{Experimentally, } K_\theta = 0.7, \qquad K_\omega = 1.5 \implies \text{Best Response}$$

Implementation of the Control Law

Regulator Implementation

For regulation tasks where the primary goal is to drive the system states to zero ($\theta = 0$), the implementation of the state-feedback control law is given by

$$\text{State-Feedback Control Law} \implies u = -0.7\theta - 0.2824\dot{\theta}$$

$$\text{Regulator: } \theta = 0$$

Tracking Implementation

To extend the controller for reference tracking applications, the control law is adjusted to operate on the tracking error vector by comparing the actual states to the desired reference angle

$$\text{For Tracking} \implies u = -\begin{bmatrix} K_\theta & K_\omega \end{bmatrix} \begin{bmatrix} \theta - \theta_{\text{ref}} \\ \dot{\theta} \end{bmatrix}$$

where the reference is set as desired

Actuation

Differential Control

The base throttle value is established at a baseline level, with differential control applied symmetrically to the actuators:

$$\text{Base Throttle} \implies u_0 = 1100$$

$$u_1 = u_0 + du$$

$$u_2 = u_0 - du$$

Slew-Rate Limiting

To preserve actuator longevity and system stability, the change in the pulse-width modulation (PWM) signal per execution step is bounded:

$$|\Delta\text{PWM}| \leq \text{Max. allowed Step}$$

where ΔPWM represents the changes Δu_1 and Δu_2. This functions as a:

$$\text{Limiting Step to avoid sudden spikes in PWM}$$

Saturation

The final control output commands are hard-clipped within physical operating limits:

$$\text{Saturation Boundary} \implies [1000, 1300]$$

State-Feedback Integration

The differential control effort du is derived via the state-feedback tracking law, balancing correction and physical damping:

$$du = -\left(K_\theta \times (\theta - \theta_{\text{ref}}) + K_\omega \times x_\omega\right)$$

where the component terms are allocated to specific control actions:

$$K_\theta \times (\theta - \theta_{\text{ref}}) \implies \text{Corrects angle}$$

$$K_\omega \times x_\omega \implies \text{Damps motion}$$

System's Block Diagram

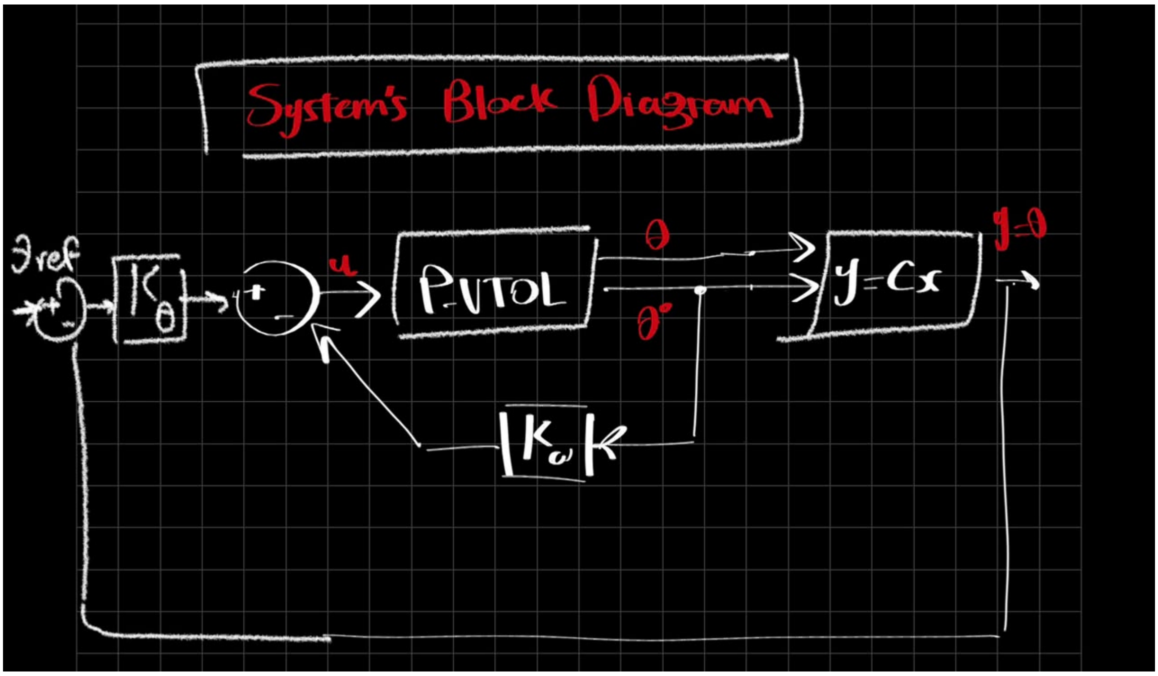

Figure 6: System's Block Diagram

labView Control Design

Following is the VI for representing the system equation, obtaining uncontrolled and controlled dynamic response through root locus, obtaining gains from desired characteristics, and plotting response before and after applying control.

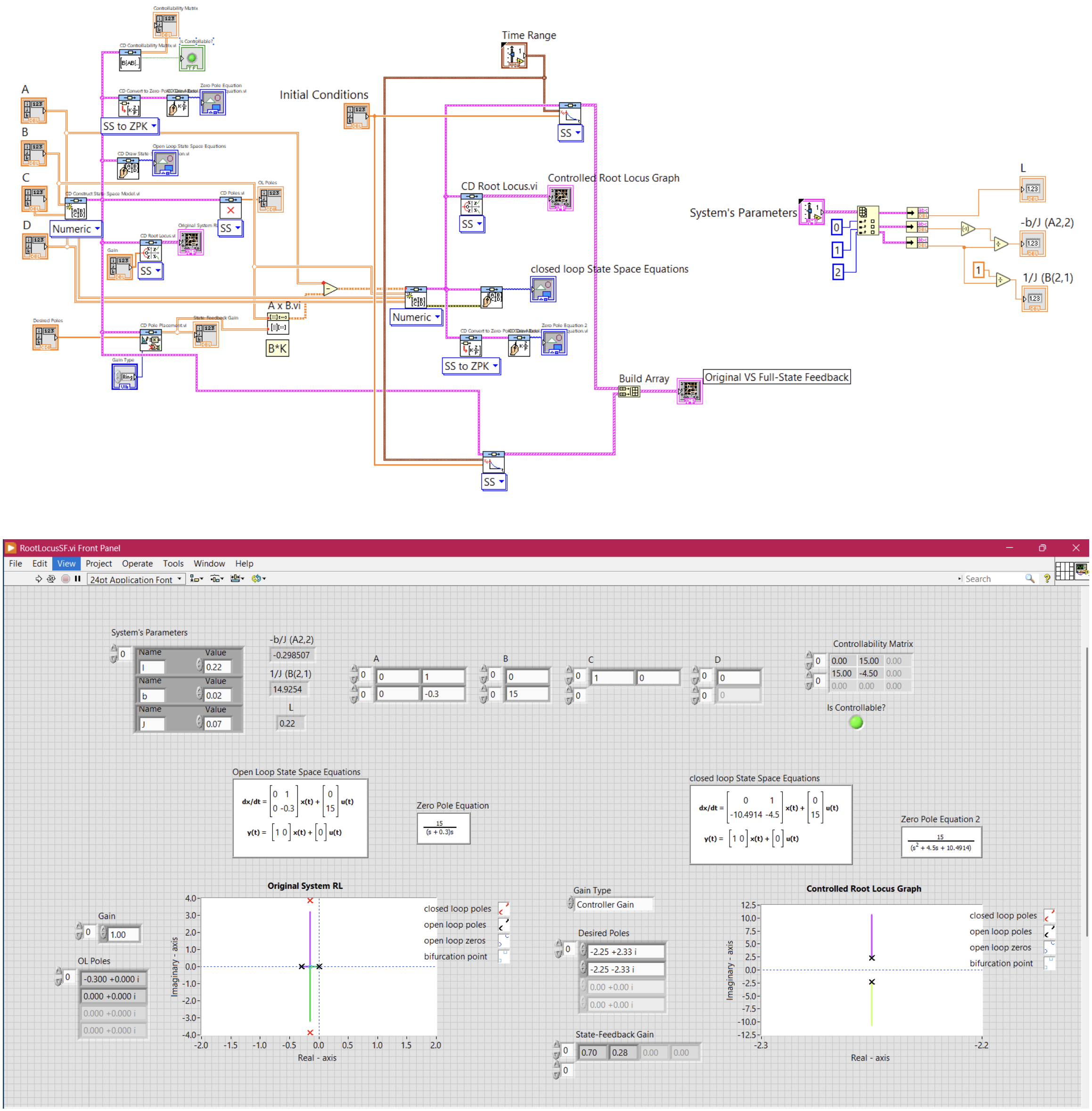

Figure 7: System Representation and Dynamic Analysis

Data Acquisition and Communication Interface

NI-VISA was instrumental in exchanging data between the Arduino and LabVIEW via serial communication. The data protocol is structured around specified write commands and read parameters.

Write Configuration

The interface transmits reference values and tunable controller feedback gains from LabVIEW to the hardware controller:

$$\text{Write} \implies \theta_{\text{ref}}, \quad K_\theta, \quad K_\omega$$

Read and Telemetry Configuration

The interface continuously streams state estimations and active control variables back to LabVIEW for real-time visualization and data logging:

$$\text{Read} \implies u_1, \quad u_2, \quad \theta, \quad \omega \qquad \text{(for Plotting)}$$

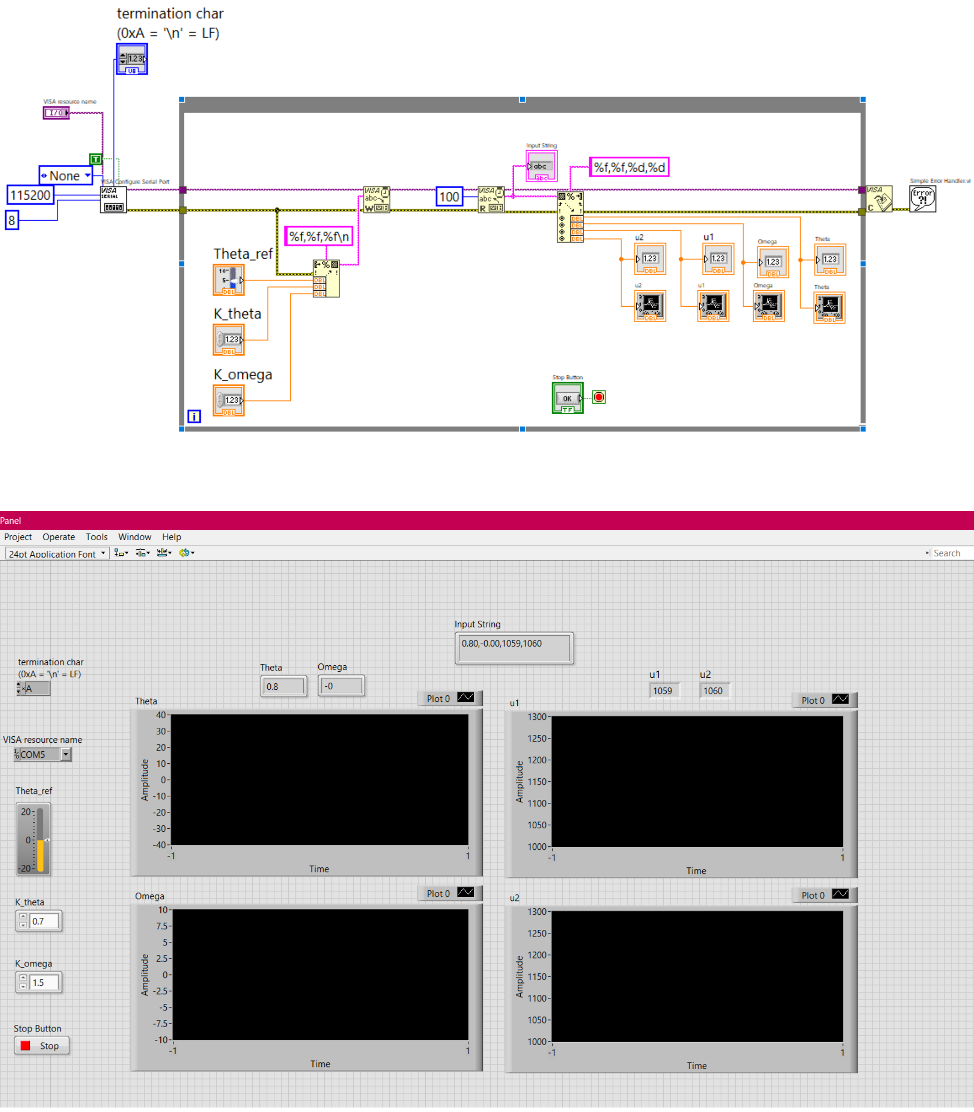

Figure 8: User Interface for running test

Experimental Results

After completing the implementation, we conducted several experiments to test the system. The state-feedback controller successfully stabilized the beam and held it at the desired equilibrium position.

During the tests, LabVIEW and the Arduino worked together in real time. LabVIEW sent the target angles and controller gains down to the Arduino. The Arduino then processed the sensor measurements, ran the control loop, adjusted the motors, and streamed the updated states and actuator efforts back to LabVIEW. This continuous data loop ran smoothly until a stop was requested. The successful balancing performance and real-time tracking are fully captured in the attached experiment video. running experiment video

Conclusion and Future Work

This project demonstrated the complete development of a P-VTOL control system starting from mechanical design and ending with real-time implementation.

Future students can extend this work by implementing:

- LQR and LQG control
- Kalman filtering for more accurate measurements
- Fuzzy Logic Control
- Model Predictive Control

References

- "PVTOL Exact Linearization Control," *IEEE Transactions on Automatic Control*, for underlying plant modeling mechanics.
- "Arduino-Based Embedded Systems Interfacing, Simulation, and LabVIEW GUI," for structural insights regarding VISA buffer configurations.
- "MPU-6050 6-Axis Accelerometer/Gyroscope Datasheet," InvenSense Inc., detailing calibration constants and LSB scaling ratios.
- "1-DOF Helicopter Stabilization Project Template Guide," Acknowledgable work done by a colleague, preceding semester.